

Interference with other measurement equipment. While ioniser equipment are sufficiently shielded to prevent any electromagnetic interference with other equipment, it needs to be verified and confirmed that any measurement values are not affected due to the operation of ESD-control ionisers.

Decay time

Static decay time is a critical performance parameter for ESD-control ionisers. This is the time required to reduce electrostatic voltage to a defined lower voltage level. Typically, a 152mm x 152mm (6-inch x 6-inch) metal plate with total capacitance of 20pF is charged to a defined potential. It is then placed at a defined distance from the air ioniser. A high-impedance voltmeter is used to monitor voltage decay on the plate as a function of time. This method is popularly called charge-plate verification mechanism.

Typically, the defined voltage, which is the possible final residual

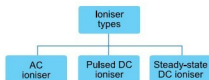


Fig. 2: Types of ESD-control ionisers

voltage on the work-surface being used for activity, should be sufficiently in the safe zone for the class of component being handled.

Problems faced

An ESD-control ioniser needs to be periodically verified for expected performance and calibrated at minimum manufacturer-recommended calibration frequency.

The ioniser has sharp emitter points that need to be cleaned of any contamination since this changes the performance drastically.

After every maintenance, it is also recommended to verify charge balance and decay time, which sometimes require significant time to com-

Periodic verification of ESD-control ionisers

ESD Association in the USA has the following standard for periodic verification of ESD-control ionisers:

ANSI/ESD SP3.3-2012: ESD Association Standard Practice for the Protection of Electrostatic Discharge Susceptible Items – Periodic Verification of Air Ionisers

plete, hence resulting in a disruption in the manufacturing process flow. Therefore any maintenance or related task on the ESD-control ioniser needs to be planned in advance.

Conclusion

With highly-sensitive ESDs coming into main-stream manufacturing, ESD-control ionisers have become inevitable for necessary process control and any type of failure prevention. Ionisers are still considered as a secondary protection mechanism. Their use does not satisfy the requirements for an EPA. **EFY**

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